

ASRA-NFM

Learning World Dynamics Through Intervention, Explicit Mechanism Discovery, and Planning Under Uncertainty

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Central question. Can an agent learn unfamiliar environments efficiently by treating actions as scientific interventions, maintaining competing explanations, and explicitly modeling the mechanisms that produce observed changes?

Motivation

- Pattern recognition may predict without discovering mechanisms.
- Purely symbolic systems are brittle under uncertain perception.
- Passive learning is inefficient in novel interactive settings.
- Agents must learn action meaning, goals, mechanisms, and plan under incomplete knowledge.

NEURAL SYMBOLIC ACTIVE EXPERIMENT

Foundational unit — Before → Action → After

$$\mathbf{T}_t = (\mathbf{s}_t, \mathbf{a}_t, \mathbf{s}_{t+1}) \quad \cdot \quad \Delta_t = \text{Diff}(\mathbf{s}_t, \mathbf{s}_{t+1})$$

$\mathbf{s}_t$ structured state · $\mathbf{a}_t$ intervention · $\mathbf{s}_{t+1}$ result · Δ_t observed effect.

Transitions evidence action semantics, rules, mechanisms, invariants, goals, and terminals.

Fig. 1 — Transition evidence unit for mechanism learning.

Object-centric state

Entities + relations — not only pixels or a global novelty vector.

Object: id, type, color, shape, position, attributes, history
Relations: adjacent · inside · aligned · blocking · approaching · connected

$$\text{Effect}(\mathbf{a}, \mathbf{o}_i, \mathbf{r}_{ij}, \mathbf{c}_t) \rightarrow \Delta \mathbf{o}_i$$

Objects/relations influence hypotheses, prediction, experiment choice, scoring, and planning.

Fig. 2 — Object–relation binding of action effects.

Architecture

ASRA-NFM architecture flowchart

Fig. 3 — Neuro-symbolic loop with memory feedback. Neural (cyan), symbolic (amber), evidence (green), planning (violet).

Tabby — Questions

Surfaces uncertainty; proposes discriminating interventions.
Functional module — not necessarily a separate LM.

Grubby — Evidence

Answers from transitions: support, contradictions, invariants, counterexamples.

Sam — Decisions

Explore, test a hypothesis, or execute a goal-directed plan.

Hypothesis management

$$\mathbf{H}_t = \{\mathbf{h}_1 \dots \mathbf{h}_k\} \quad \text{Score}(\mathbf{h}) = \text{Acc} + \alpha \text{Cov} - \beta \text{Cplx} - \gamma \text{Contr}$$

H1: ACTION_2 moves every red object. · H2: moves object adjacent to agent. · H3: moves selected object if path clear.

Favor accurate, covering, simple, falsification-surviving explanations.

Fig. 4 — Competing falsifiable mechanisms distinguished by intervention.

Active experimentation

$$\mathbf{a}^* = \text{argmax}_{\mathbf{a}} [\text{IG}(\mathbf{a}; \mathbf{H}_t) + \lambda \text{P}(\text{progress} | \mathbf{a}) - \mu \text{Risk}(\mathbf{a}) - \nu \text{Cost}(\mathbf{a})]$$

Actions valued for progress *or* for revealing which world-model hypothesis is correct.

Memory systems

Episodic · transition · semantic · procedural · failure · experiment (hypotheses, configs, results, conclusions).

Applications

Initial testbed: interactive ARC-AGI-style environments. Broader: novel envs, autonomous ML experimentation, scientific discovery, robotics, biological decision systems, physical AI. Competition adapters share one reasoning core.

Results

Preliminary · placeholders only

Benchmark [BENCHMARK] · Score [SCORE] · Prev [PREVIOUS_SCORE] · Δ [DELTA]
N [N] · Date [DATE] · Version [VERSION] · Hardware [HARDWARE]

Placeholder evaluation charts

Fig. 5 — Editable charts; dashed bars = unfilled measurements. No invented results.

Claim (hypothesis until validated)

An object-centric neuro-symbolic agent with explicit hypothesis management and active intervention can learn novel mechanisms more accurately and with fewer actions than passive, neural-only, and symbolic-only baselines.

Contributions

1 Transition-centered learning · 2 Separate action semantics / mechanisms / goals · 3 Object-centric causality · 4 Competing hypotheses · 5 Hybrid world models · 6 Information-seeking interventions · 7 Interpretable traces · 8 Reusable scientific-reasoning core